INFRARED LIGHT TRACING ROBOT

Project Artifacts & Documentation

Project objective: Design and develop an autonomous robot capable of detecting and tracing infrared (IR) light sources for navigation and task-specific applications.

Note: This document is a professional project-artifact template based on the project description available in the provided resume. Exact source code, component values, circuit files, prototype photographs, and simulation screenshots were not present in the source material, so placeholders are provided rather than invented project evidence.

1. Source Code

Upload the actual source-code files used in the project to the GitHub repository. Recommended organization:

Suggested folder: /Source_Code/

• Main program/source file
• Header/configuration files, if used
• Motor-control code
• IR-sensor reading/control code
• Any supporting libraries or configuration files

GitHub note: Do not upload code that you did not actually use in the project. Replace this section with the real source files before publishing the repository.

2. Circuit Diagram

IR Sensor(s)	→	Controller / Control Logic	→	Motor Driver	→	DC Motors
IR light detection		Processes sensor input		Drives motors		Robot movement

Insert actual circuit schematic here. The exact circuit connections and component values should be taken from the circuit actually built or simulated for this project.

3. Block Diagram

IR Light Source	→	IR Sensor	→	Controller	→	Motor Driver	→	Motors / Robot
Target		Detection		Decision & control		Power switching		Movement

Working concept: The IR sensing section detects the IR source. The controller interprets the sensor information and provides control signals to the motor driver, which drives the motors so that the robot moves toward/traces the detected IR source.

4. Project Report / PDF

Project overview: The Infrared Light Tracing Robot is an autonomous robotic system designed to detect and track infrared light sources. The project focuses on sensor-based detection, control logic, and motor actuation for automated navigation.

Applications: Automated navigation, IR-source tracking, robotics demonstrations, and sensor-based autonomous systems.

Key learning outcomes: Practical understanding of embedded-system design, IR sensing, motor control, autonomous navigation, and integration of electronic hardware.

Recommended report sections: Introduction → Objectives → Components → Circuit Design → Block Diagram → Working Principle → Software/Algorithm → Hardware Implementation → Testing → Results → Applications → Limitations → Future Scope → Conclusion.

5. Proteus / CAD Files

If the project was simulated or designed using Proteus, CAD, or another EDA tool, upload the original project files to a dedicated repository folder.

Suggested folder: /Simulation_or_CAD/

Include the actual project file(s), schematic files, PCB files if applicable, and any libraries required to open the design.

Status: The provided resume does not specify that Proteus/CAD was used for this project, so no specific tool is claimed here.

6. Photos of Working Prototype

PHOTO 1 — FRONT VIEW	PHOTO 2 — TOP VIEW
Insert actual prototype photograph here	Insert actual prototype photograph here
PHOTO 3 — IR SENSOR / CONTROL SECTION	PHOTO 4 — ROBOT IN OPERATION
Insert actual photograph here	Insert actual photograph here

7. Simulation Screenshot

Insert a clear screenshot of the actual simulation, showing the circuit and relevant simulated behavior/results.

SIMULATION SCREENSHOT
Insert actual simulation screenshot here

GitHub repository structure

Infrared-Light-Tracing-Robot/
■■■ README.md
■■■ Source_Code/
■■■ Circuit_Diagram/
■■■ Block_Diagram/
■■■ Project_Report/
■■■ Simulation_or_CAD/
■■■ Prototype_Photos/
■■■ Simulation_Screenshots/

Final step: Replace all placeholders with your real project artifacts before making the GitHub repository public. This keeps the repository accurate and avoids presenting generated diagrams, photos, or files as actual project evidence.